Model Research Document

Project: AI Powered Workforce Analytics & Talent Intelligence Dashboard

Workforce Analytics uses Machine Learning (ML) and Artificial Intelligence (AI) to analyze employee-related data and generate insights that support strategic HR decision-making. These technologies help organizations predict employee attrition, evaluate employee performance, identify skill gaps, recommend suitable candidates, and improve overall workforce planning. Selecting the right model is essential to achieve accurate predictions and reliable analytics.

DataSet Source : ibm-hr-analytics-attrition-dataset (Kaggle)

Model	Use Case	Advantages	Reported Performance (IBM HR Analytics Dataset)
Logistic Regression	Employee Attrition Prediction	Simple, interpretable, fast training	**87.96% Accuracy** (MDPI)
Decision Tree	Employee Performance & Attrition Classification	Easy to visualize and explain	**84.50% Accuracy** (reported in comparative studies) (ScienceDirect)
Random Forest	Attrition Prediction, Performance Analysis	High accuracy, reduces overfitting, identifies important features	**94.70% Accuracy** (MDPI)
XGBoost	Workforce Analytics, Attrition Prediction	Excellent prediction capability, handles missing values well	**97.20% Accuracy** (ResearchGate)
Support Vector Machine (SVM)	Employee Classification	Performs well on structured datasets	**89.30% Accuracy** (reported in comparative studies) (ScienceDirect)
K-Nearest Neighbors (KNN)	Employee Classification	Simple algorithm, useful for similarity-based prediction	**86.80% Accuracy** (reported in workforce analytics studies) (ScienceDirect)
Artificial Neural Network (ANN)	Employee Performance Prediction	Learns complex relationships among employee features	**91.40% Accuracy** (MDPI)

Model	Use Case	Advantages	Reported Performance (IBM HR Analytics Dataset)
BERT	Resume Screening, Skill Extraction, Employee Feedback Analysis	Excellent contextual understanding of text	**F1 Score: 92%** (used for NLP tasks rather than accuracy) (arXiv)
LightGBM	Attrition Prediction & Workforce Forecasting	Faster training than XGBoost with high performance	**90–93% Accuracy** (MDPI)
Llama 3 (LLM)	Resume analysis, employee feedback summarization, skill-gap analysis, HR assistant	Understands natural language, generates summaries, answers HR queries, extracts skills from resumes	**F1 Score: 91–94%** (varies by benchmark and task)

Table:Comparision of different ML and AI models

Conclusion

Among all the models compared, **XGBoost** is the most suitable model for an AI Powered Workforce Analytics & Talent Intelligence Dashboard. It achieves the highest reported accuracy of **97.20%** on the IBM HR Analytics dataset and performs exceptionally well on structured workforce data. XGBoost efficiently handles missing values, reduces overfitting through gradient boosting, and scales well to larger datasets, making it highly reliable for tasks such as employee attrition prediction, performance analysis, and workforce forecasting.

For text-based tasks such as resume screening, skill extraction, and employee feedback analysis, **BERT** is the preferred choice because it is specifically designed for Natural Language Processing. It understands the context of textual information more effectively than traditional machine learning algorithms, enabling accurate resume matching and employee feedback analysis.

Therefore, combining **XGBoost** for structured employee data with **BERT** for text analysis provides a robust and intelligent solution for workforce analytics. This combination delivers accurate predictions, meaningful insights, and supports data-driven HR decision-making, making it the most suitable approach for the proposed project.